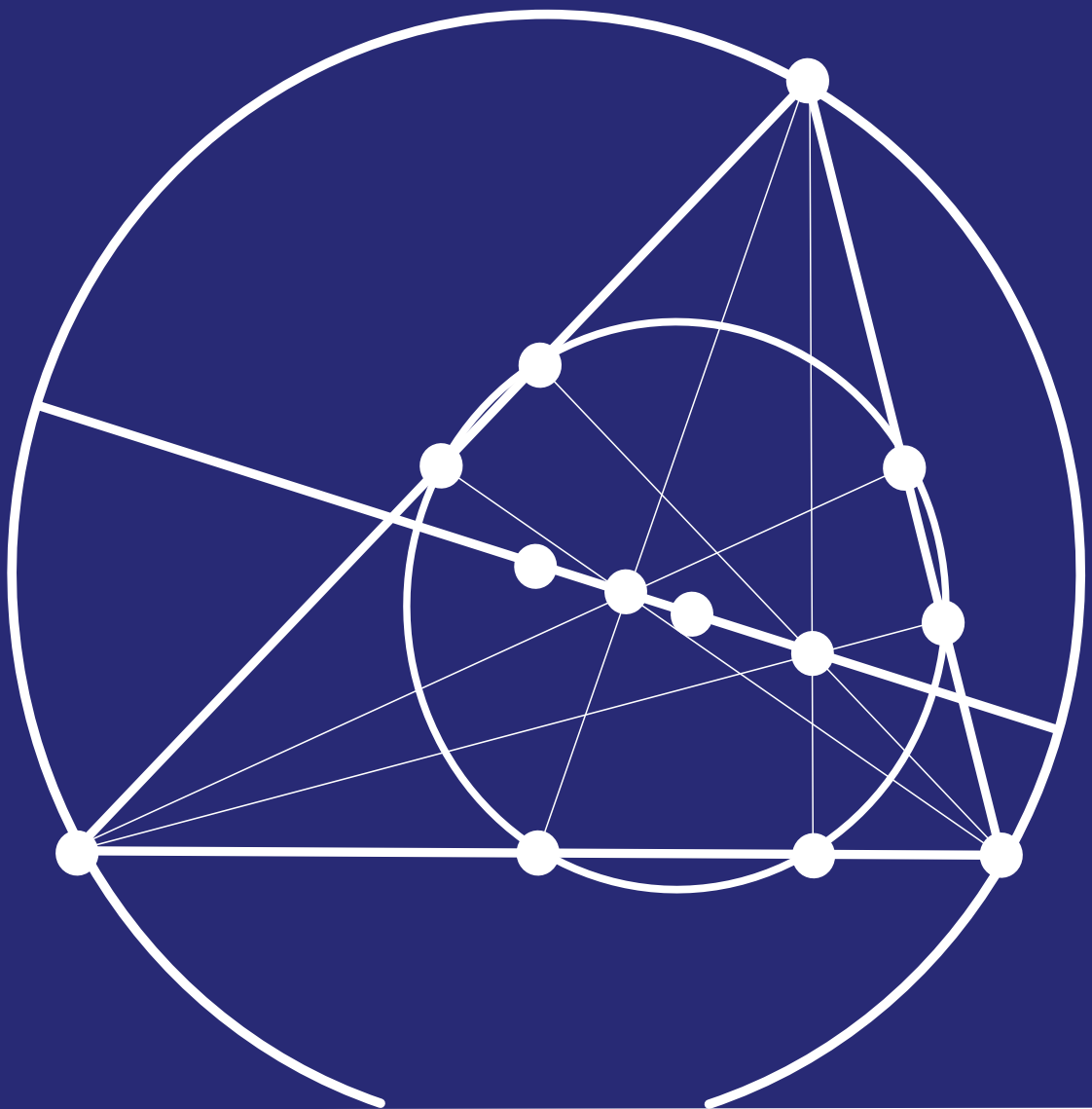
Eaman Eftekhary

Complex Numbers in Contest Geometry

Sample Content



SyMath

Preface: A Gateway to Systematic Geometry

Welcome to this curated preview of the SyMath book ***Complex Numbers in Contest Geometry***.

The traditional approach to competitive olympiad geometry often relies on "synthetic" methods—searching for highly specific auxiliary constructions and waiting for an unpredictable "aha!" moment. This handbook is built on a different philosophy: that advanced geometry can alternatively be approached as a structured, predictable, and highly systematic algebraic algorithm through the language of complex numbers.

This short booklet is an extracted sample of ***Complex Numbers in Contest Geometry***, designed to show you both the foundational mechanics and the elite competitive power of this toolkit. Inside, you will find:

- **Foundational Toolkits:** Two elementary chapters introducing the core vocabulary of the complex plane.
- **Advanced Apparatus:** Two chapters detailing deep classical theorems and structural transformations.
- **The Framework in Action:** Four medium-difficulty IMO problems, complete with fully executed, systematic complex-number solutions.

Whether you are a national team aspirant, an Olympiad coach, or an advanced mathematics enthusiast, we hope these select pages demonstrate how a unified algebraic perspective can turn geometric complexity into absolute computational clarity.

Build your toolkit. Solve with confidence.

Eaman Eftekhary
School of Mathematics, IPM

To explore the complete, uninterrupted manual containing the full approach, a much larger variety of techniques and major theorem, all 50 fully solved IMO geometry problems (1994–2025) and over 250 advanced exercises, visit the official SyMath library at: **symath.gumroad.com**

Chapter 1 Beyond Reality

The Complex Plane

Negative real numbers do not admit square roots among real numbers. Instead of abandoning the search for such roots, one may enlarge the notion of a number. In fact, we may adjoin a formal square root i of $-1 \in \mathbb{R}$ to the set of real numbers and explore the consequences in this extended system. The imaginary number i satisfies $i^2 = -1$. It is crucial to understand that i is not *the square root of -1* in the same sense that $\sqrt{4} = 2$, because there are actually two square roots of -1: i and $-i$. The symbol i is defined to be one of them, and once we choose i, the other is necessarily $-i$. Real multiples of i, of the form iy for $y \in \mathbb{R}$, must of course be included as well. Furthermore, the sum of any $x \in \mathbb{R}$ and any imaginary number iy (with $y \in \mathbb{R}$) is also a number in our new field. Remarkably, this construction already yields an extremely rich set of numbers—the complex numbers—with profound properties and applications, defined as

$$\mathbb{C} = \{x + iy \mid x, y \in \mathbb{R}\} \simeq \{(x, y) \mid x, y \in \mathbb{R}\} = \mathbb{R}^2. \tag{1.1}$$

In other words, a complex number is a formal combination $x + iy$, where x and y are real numbers, and may be identified with the pair $(x, y) \in \mathbb{R}^2$. As mentioned in the previous section, this geometric identification was first suggested by Gauss [Gau31]. From this perspective, complex numbers may be regarded as points (or vectors) in the plane, and conversely, every point (or vector) in the Euclidean plane corresponds to a complex number.

This identification $\mathbb{C} \simeq \mathbb{R}^2$ is more than just a convenient correspondence—it is an isomorphism of real vector spaces. This means that as far as addition and scalar multiplication by real numbers are concerned, complex numbers behave exactly like vectors in the plane. However, complex numbers have an additional operation: multiplication, which gives $\mathbb{C}$ a richer algebraic structure than $\mathbb{R}^2$.

Real numbers embed naturally into $\mathbb{C}$. Indeed, if $r \in \mathbb{R}$ is a real number, then $r + i \cdot 0$ is the corresponding complex number, which we simply denote by r. We write $\mathbb{R} \subset \mathbb{C}$ for this natural inclusion. Geometrically, $\mathbb{R}$ corresponds to the first axis in the Euclidean plane $\mathbb{R}^2$, called the real axis. The second axis, corresponding to the set

$$i\mathbb{R} = \{iy \mid y \in \mathbb{R}\} \subset \mathbb{C},$$

is called the imaginary axis.

Conjugation

Every complex number $a = x + iy \in \mathbb{C}$ has a conjugate

$$\overline{a} = x - iy = x + i(-y) \in \mathbb{C}.$$

Geometrically, the conjugate is obtained by reflecting a across the real axis $\mathbb{R} \subset \mathbb{C}$. Figure 1.1

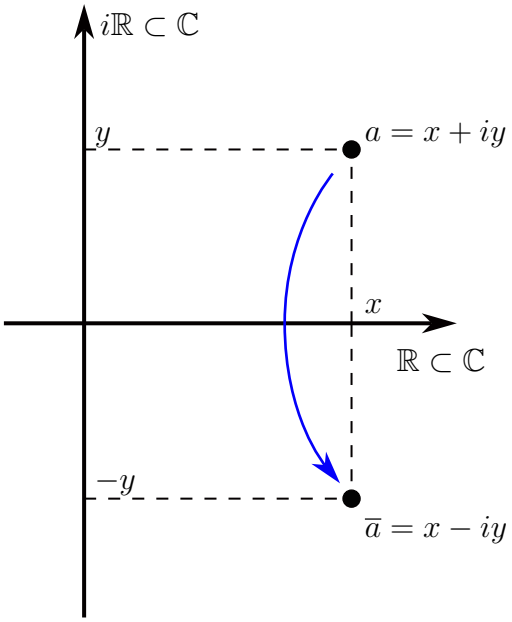


Figure 1.1: Complex conjugation is the reflection of a point across the real axis

illustrates this process. Notice that conjugation is an involution:

$$\overline{\overline{a}} = a \qquad \forall\, a \in \mathbb{C}.$$

This operation preserves addition and multiplication in a special way, as we will see in exercises. Note that both a and $\overline{a}$ correspond to vectors in $\mathbb{R}^2$, and hence we may define their lengths, called the modulus or complex norm (or simply, the norm) $|a| \in \mathbb{R}^{\geq 0}$. It is given by

$$|a| = |\overline{a}| := \sqrt{x^2 + y^2}, \qquad \forall\, a = x + iy \in \mathbb{C}. \tag{1.2}$$

The modulus of a complex number extends the concept of absolute value from real numbers. For a real number x, viewed as $x + 0i$, we have

$$|x| = \sqrt{x^2} = |x|.$$

Note that the modulus of a complex number $a \in \mathbb{C}$ is always non-negative, and equals zero if and only if $a = 0$ (i.e. a is the origin in $\mathbb{R}^2$).

Moreover, if $a = x + iy$ and $b = x' + iy'$ are two complex numbers corresponding to two points in the complex plane, then

$$|a - b| = \sqrt{(x - x')^2 + (y - y')^2}$$

is precisely the Euclidean distance (or simply, the distance) between a and b. Equivalently, it is the Euclidean length (or simply, the length) of the line segment connecting a to b.

It is often convenient to avoid writing a complex number $a = x + iy$ explicitly as the sum of its real part x and imaginary part iy. Instead, the conjugate $\overline{a}$ provides the necessary information, since

$$\mathrm{Re}(a) = x = \frac{a + \overline{a}}{2} \quad \text{and} \quad \mathrm{Im}(a) = y = \frac{a - \overline{a}}{2}. \tag{1.3}$$

These formulas are particularly useful because they express the real and imaginary parts purely in terms of a and $\overline{a}$ without explicitly decomposing a into $x + iy$. This is the first hint that conjugation is not just a geometric operation but also a powerful algebraic tool.

Chapter 2 The VIP Circle

The study of complex numbers becomes particularly rich when we focus on those numbers whose modulus equals one. These numbers lie on the unit circle in the complex plane—a simple geometric object that unexpectedly reveals deep connections between algebra, geometry, and number theory.

The Unit Circle in Complex Plane

The set of complex numbers with modulus 1 is called the unit circle, denoted by

$$\mathbb{S} = \{z \in \mathbb{C} \mid |z| = 1\} \subset \mathbb{C}.$$

Geometrically, these numbers correspond to the points on the circle of radius 1 centered at the origin. The set of complex numbers with modulus smaller than 1 is denoted by $\mathbb{D}$, and the union of $\mathbb{D}$ and $\mathbb{S}$ is denoted by $\overline{\mathbb{D}}$:

$$\mathbb{D} = \{z \in \mathbb{C} \mid |z| < 1\} \quad \text{and} \quad \overline{\mathbb{D}} = \{z \in \mathbb{C} \mid |z| \leq 1\} = \mathbb{D} \cup \mathbb{S}.$$

The unit circle is arguably the most important subset of the complex plane. It serves as a bridge between algebra (through complex multiplication) and geometry (through rotations), and it sets a natural setting for trigonometry. Many of the most beautiful properties of complex numbers emerge when we restrict our attention to the unit circle.

Every point $a = x + iy$ on the unit circle is completely determined by the angle θ between the corresponding vector and the positive real axis[1]. This angle, measured in radians, is called the argument of a, and is denoted by $\arg(a)$. We assume that $\arg(a)$ takes its values in the interval $(-\pi, \pi]$.

From basic trigonometry, we know that for such a point,

$$x = \cos\theta \quad \text{and} \quad y = \sin\theta,$$

as illustrated in Figure 2.1. Thus any point on the unit circle is of the form

$$a_\theta = \cos\theta + i\sin\theta, \quad \text{with } \theta \in \mathbb{R}.$$

Notice that a_θ and $a_{\theta+2\pi}$ represent the same point, since adding 2π to the angle corresponds to making a full revolution around the circle.

The algebraic structure of the unit circle reveals itself when we multiply two such numbers. Let

$$a_\alpha = \cos\alpha + i\sin\alpha \quad \text{and} \quad a_\beta = \cos\beta + i\sin\beta$$

be two points on $\mathbb{S}$. Their product is:

$$\begin{aligned} a_\alpha a_\beta &= \big(\cos(\alpha) + i\sin(\alpha)\big)\big(\cos(\beta) + i\sin(\beta)\big) \\ &= \big(\cos(\alpha)\cos(\beta) - \sin(\alpha)\sin(\beta)\big) \\ &\quad + i\big(\cos(\alpha)\sin(\beta) + \sin(\alpha)\cos(\beta)\big) \\ &= \cos(\alpha+\beta) + i\sin(\alpha+\beta) \\ &= a_{\alpha+\beta}. \end{aligned} \tag{2.1}$$

[1] As mentioned in the first section of this chapter, the term *argument* was first suggested by Cauchy.

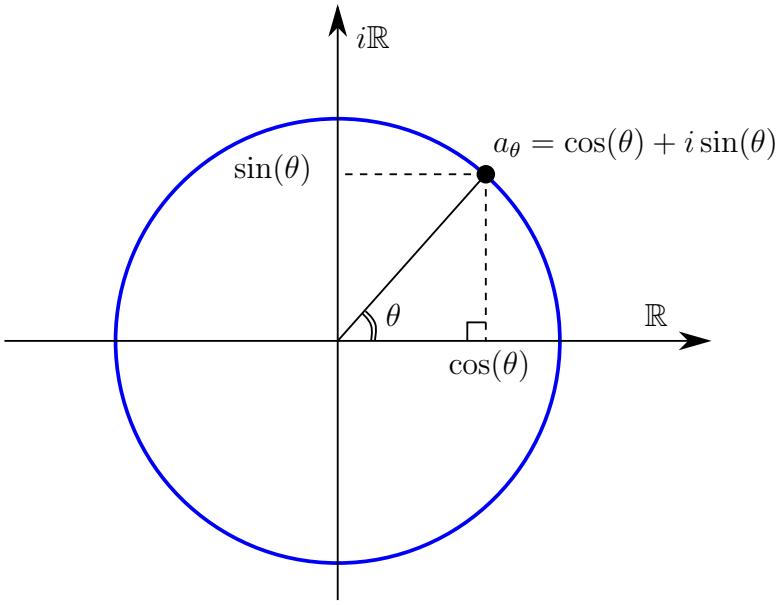


Figure 2.1: Every point on the unit circle is of the form $a_\theta = \cos(\theta) + i\sin(\theta)$ for some $\theta \in \mathbb{R}$.

This is a *remarkable formula*! The exponential notation $e^{i\theta}$ for a_θ is justified by (2.1), since the outcome of the latter computation may be formulated as

$$e^{i(\alpha+\beta)} = e^{i\alpha}e^{i\beta} \qquad \forall\, \alpha, \beta \in \mathbb{R}.$$

There is of course much more depth in this *convention*, but we will keep our explanation limited to the above multiplication property and refer the interested reader to the second volume of this book for further explanation. Note that Euler's formula is a particular case of the formula

$$e^{i\theta} = \cos\theta + i\sin\theta. \tag{2.2}$$

Setting $\theta = \pi$ in this formula implies

$$1 + e^{\pi i} = (1 + \cos\pi) + \sin\pi = 1 - 1 = 0, \tag{2.3}$$

which is Euler's famous formula.

Equation (2.1) tells us that multiplying complex numbers on the unit circle corresponds to adding angles (arguments). It also means that we can represent rotations algebraically: multiplying by $a_\theta = e^{i\theta}$ rotates any complex number by angle θ counterclockwise about the origin. For example:

- Multiplication by
$$i = \cos\left(\frac{\pi}{2}\right) + i\sin\left(\frac{\pi}{2}\right)$$
rotates the complex plane by $90°$.
- Multiplication by $-1 = \cos\pi + i\sin\pi$ corresponds to a rotation by $180°$, which is the same as reflection with respect to the origin.
- Multiplication by
$$\frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2} = \cos\left(\frac{\pi}{4}\right) + i\sin\left(\frac{\pi}{4}\right)$$
rotates everything by $45°$.

This geometric interpretation of multiplication extends to all complex numbers, not just those on

the unit circle. For any complex number $a = |a|a_\theta$, multiplication by a scales by $|a|$ and rotates by θ. This is why complex multiplication is so geometrically natural.

Exercise 2.1

Let $z = 1 + i$. Compute iz and $(-1)z$, then plot these three numbers in the complex plane. Verify geometrically that multiplication by i rotates z by $90°$ counterclockwise, while multiplication by -1 rotates z by $180°$.

Exercise 2.2

Use the group structure of the unit circle to show that for any integer n,

$$(\cos\theta + i\sin\theta)^n = \cos(n\theta) + i\sin(n\theta) \quad \textit{(de Moivre's formula)}.$$

Use this to find formulas for $\cos(3\theta)$ and $\sin(3\theta)$ in terms of $\cos\theta$ and $\sin\theta$.

The Roots of Unity

Among the most important points on the unit circle are those that correspond to dividing the circle into equal arcs. These points are called roots of unity. Roots of unity are fundamental in many areas of mathematics. They appear in number theory (cyclotomic fields), algebra (group theory), geometry (regular polygons), and even in applied mathematics (Fast Fourier Transform). Their study bridges seemingly disconnected areas of mathematics.

Definition 2.1

For a positive integer n, an nth root of unity is a complex number z such that $z^n = 1$.

If we set

$$\zeta_n = \cos\left(\frac{2\pi}{n}\right) + i\sin\left(\frac{2\pi}{n}\right) = e^{2\pi i/n},$$

then the nth power of ζ_n is the identity:

$$\zeta_n^n = \left(e^{2\pi i/n}\right)^n = e^{2\pi i} = 1.$$

Moreover, the first n powers of ζ_n, i.e. the complex numbers

$$\zeta_n^k = e^{2\pi ik/n}, \quad \text{for } k = 0, 1, 2, \ldots, n-1,$$

are distinct and satisfy

$$(\zeta_n^k)^n = e^{2\pi ik} = 1.$$

One can see that these numbers are different from one another since their arguments are the evenly distributed (and different) angles

$$0, \frac{2\pi}{n}, \frac{4\pi}{n}, \cdots, \frac{2(n-1)\pi}{n} \in [0, 2\pi).$$

These give n solutions to $z^n = 1$:

$$1,\ \zeta_n,\ \zeta_n^2,\ \ldots,\ \zeta_n^{n-1}$$

Exercise 2.3

Prove that every nth root of unity (i.e. every point z satisfying $z^n = 1$) is of the form ζ_n^k for some $k = 0, 1, \ldots, n-1$.

Geometrically, these n points are equally spaced around the unit circle, forming the vertices of a regular n-gon inscribed in the circle. We will denote the set of nth roots of unity by U_n. Therefore,

$$U_n = \{\zeta_n^k \mid k = 0, 1, \ldots, n-1\} \subset \mathbb{S} \subset \mathbb{C}.$$

This cyclic structure has beautiful algebraic and geometric properties. One of the most useful ones is that the sum of all nth roots of unity is zero for $n > 1$:

Lemma 2.1

For every integer $n > 0$, the sum of nth roots of unity is zero, i.e.

$$1 + \zeta_n + \zeta_n^2 + \cdots + \zeta_n^{n-1} = 0.$$

Proof. Note that $z^n - 1$ factors as

$$z^n - 1 = (z-1)(1 + z + z^2 + \cdots + z^{n-1}).$$

Therefore, for every complex number $z \neq 0$ we have

$$1 + z + z^2 + \cdots + z^{n-1} = \frac{z^n - 1}{z - 1}.$$

In particular, if we set $z = \zeta_n$ we find

$$1 + \zeta_n + \zeta_n^2 + \cdots + \zeta_n^{n-1} = \frac{\zeta_n^n - 1}{\zeta_n - 1} = \frac{1 - 1}{1 - \zeta_n} = 0.$$

This completes the proof of the lemma. □

Chapter 3 The Euler Line and The Nine-Point Circle

Euler Line

Our first observation is that the circumcenter, centroid, and orthocenter are collinear:

Theorem 3.1

For every triangle $\triangle ABC$, the circumcenter O, centroid M, and orthocenter H of $\triangle ABC$ are collinear. Moreover, M lies between O and H on the line containing these three points, called the Euler line of $\triangle ABC$, while the length of MH is twice the length of OM.

Proof. After appropriate translation and scaling, we may assume that the vertices A, B and C of $\triangle ABC$ lie on the unit circle and correspond to complex numbers $a, b, c \in \mathbb{S} \subset \mathbb{C}$, respectively. Let

$$o = o_{a,b,c}, \quad m = m_{a,b,c} \quad \text{and} \quad h = h_{a,b,c}$$

denote the complex numbers associated with O, M and H, respectively. Then o is the origin $0 \in \mathbb{C}$. On the other hand, our previous computations imply

$$m = \frac{a+b+c}{3} \quad \text{and} \quad h = a+b+c = 3m. \tag{3.1}$$

In other words, the vector $\vec{OH}$ equals $3\vec{OM}$. This observation implies all claims in the statement of the theorem. □

Exercise 3.1

If H is the orthocenter of $\triangle ABC$, prove that the Euler lines of $\triangle ABC, \triangle HAB, \triangle HBC$ and $\triangle HCA$ are concurrent.

The Nine-Point Circle

Let us continue to assume that the vertices of triangle $\triangle ABC$ lie on the unit circle and correspond to complex numbers

$$a, b, c \in \mathbb{S}.$$

Further assume that H_A, H_B and H_C are the feet of the altitudes, M_A, M_B and M_C are the midpoints of the sides, H is the orthocenter, and M is the centroid of $\triangle ABC$. Denote the corresponding complex numbers by

$$h_a, \quad h_b, \quad h_c, \quad m_a, \quad m_b, \quad m_c, \quad h \text{ and } m,$$

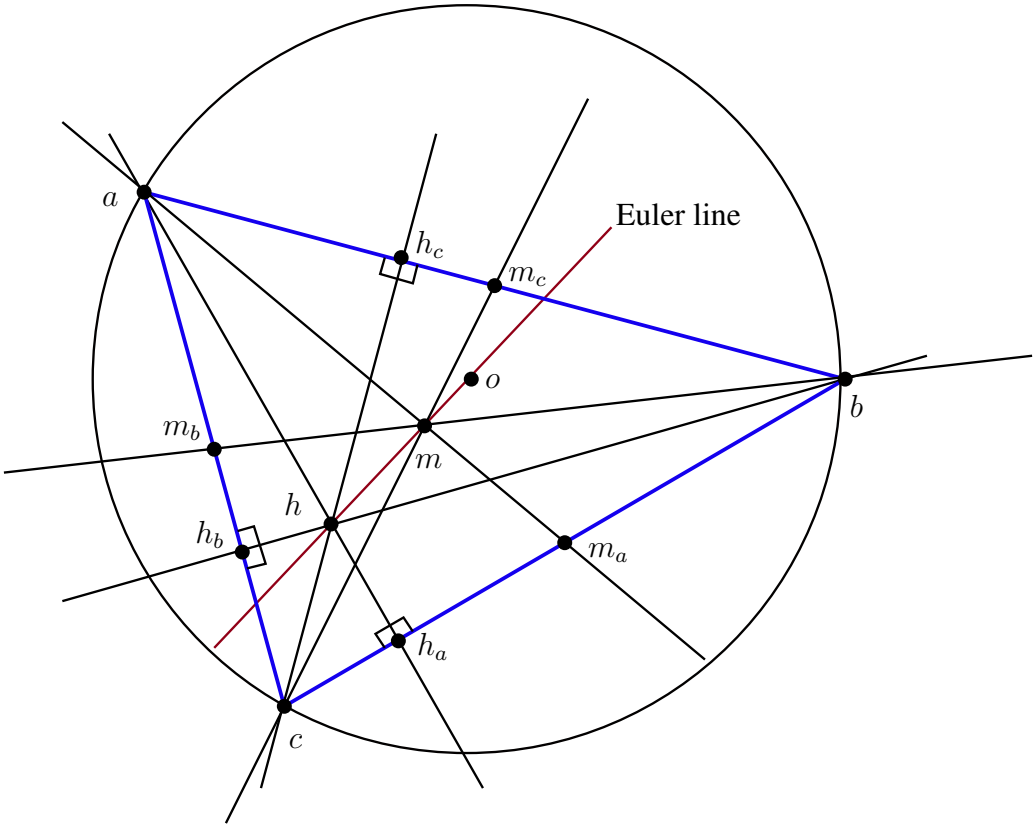


Figure 3.1: The Euler line of the triangle with vertices a, b, c includes its circumcenter $o = o_{a,b,c}$, its centroid $m = m_{a,b,c}$, and its orthocenter $h = h_{a,b,c}$.

respectively. Therefore,

$$m_a = \frac{b+c}{2}, \quad m_b = \frac{c+a}{2} \quad \text{and} \quad m_c = \frac{a+b}{2}.$$

Furthermore, earlier considerations imply that

$$m = \frac{a+b+c}{3} \quad \text{and} \quad h = a+b+c.$$

The intersection point $z = h_a$ of $\mathcal{L}_{a,h_a}$ and $\mathcal{L}_{b,c}$ satisfies the following two equalities:

$$\begin{aligned} &(1) \quad z + bc\overline{z} = b + c \qquad \text{and} \\ &(2) \quad \Big(\frac{1}{b} - \frac{1}{c}\Big) z + (b-c)\overline{z} = \Big(\frac{1}{b} - \frac{1}{c}\Big) a + (b-c)\frac{1}{a}. \end{aligned} \tag{3.2}$$

Equation (2) may be rewritten as

$$z - bc\overline{z} = a - \frac{bc}{a}.$$

Together with equation (1) from (3.2), this equality implies

$$2z = (a+b+c) - \frac{bc}{a}.$$

Repeating the same argument for the other altitudes gives the feet of the altitudes:

$$\begin{aligned} h_a &= \frac{a+b+c}{2} - \frac{bc}{2a}, \\ h_b &= \frac{a+b+c}{2} - \frac{ca}{2b} \quad \text{and} \\ h_c &= \frac{a+b+c}{2} - \frac{ab}{2c}. \end{aligned} \tag{3.3}$$

Lemma 3.1

Under the above assumptions, the circumcenter of triangle $\triangle M_A M_B M_C$ is given by the complex number $\frac{a+b+c}{2}$.

Proof. By the earlier computation of the circumcenter we have

$$\begin{aligned} o_{m_a,m_b,m_c} &= \Big(\sum\nolimits^{\circ} |m_a|^2(m_b - m_c)\Big) \Big/ \Big(\sum\nolimits^{\circ} \overline{m_a}(b-c)\Big) \\ &= \Big(\sum\nolimits^{\circ} \frac{(b+c)^2}{4bc} \cdot \frac{c-b}{2}\Big) \Big/ \Big(\sum\nolimits^{\circ} \frac{(b+c)}{2bc} \cdot \frac{c-b}{2}\Big) \\ &= \frac{1}{2}\Big(\sum\nolimits^{\circ} a(b+c)^2(c-b)\Big) \Big/ \Big(\sum\nolimits^{\circ} a(b+c)(c-b)\Big) \\ &= \frac{1}{2}\Big(\sum\nolimits^{\circ} \big(a(c^3 - b^3) + abc(c-b)\big)\Big) \Big/ \Big(\sum\nolimits^{\circ} ab(a-b)\Big). \end{aligned}$$

On the other hand,

$$\begin{aligned} \sum\nolimits^{\circ} \big(a(c^3 - b^3) + abc(c-b)\big) &= \sum\nolimits^{\circ} ab(a^2 - b^2) \\ &= \sum\nolimits^{\circ} ab(a-b)(a+b+c-c) \\ &= (a+b+c)\sum\nolimits^{\circ} ab(a-b) - abc\sum\nolimits^{\circ}(a-b) \\ &= (a+b+c)\sum\nolimits^{\circ} ab(a-b). \end{aligned}$$

Therefore, it follows that

$$\begin{aligned} o_{m_a,m_b,m_c} &= \frac{1}{2}\frac{(a+b+c)\sum^{\circ} ab(a-b)}{\sum^{\circ} ab(a-b)} \\ &= \frac{a+b+c}{2}, \end{aligned}$$

completing the proof of the lemma. □

Theorem 3.2

Given a triangle $\triangle ABC$, let H_A, H_B and H_C be the feet of the altitudes, M_A, M_B and M_C be the midpoints of the sides, and H be the orthocenter of $\triangle ABC$. Then the points $H_A, H_B, H_C, M_A, M_B, M_C$ and the midpoints of the line segments AH, BH and CH are concyclic. Furthermore, the nine-point center, which is the center of the nine-point circle containing these 9 points, lies on the Euler line of $\triangle ABC$.

Proof. Let us continue to use the notation set in this section for the complex points associated with these points, keeping the assumption that the vertices of the triangle lie on the unit circle. We have

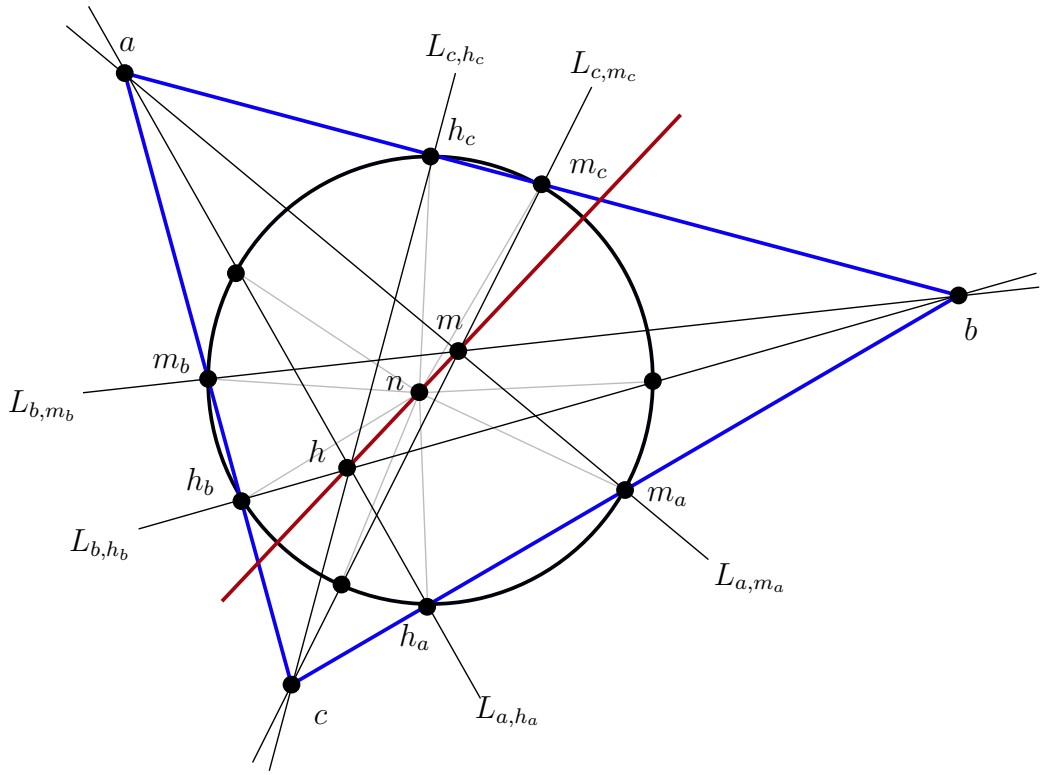


Figure 3.2: The feet of the altitudes, the midpoints of the sides, and the midpoints of the line segments connecting the vertices to the orthocenter of the triangle lie on the nine-point circle. The nine-point center n lies on the Euler line of the triangle.

already seen in Lemma 3.1 that the circumcenter N of $\triangle M_A M_B M_C$ corresponds to the complex number

$$n = o_{m_a,m_b,m_c} = \frac{a+b+c}{2}.$$

This point is clearly on the Euler line of the triangle, which consists of real multiples of $a+b+c$. We must now show that the other six points, i.e.,

$$h_a,\ h_b,\ h_c,\ \frac{a+h}{2},\ \frac{b+h}{2} \text{ and } \frac{c+h}{2},$$

also lie on the circumcircle of $\triangle M_A M_B M_C$. The radius r of the latter circle may be computed from

$$r^2 = |n-m_a|^2 = \left|\frac{a+b+c}{2} - \frac{b+c}{2}\right|^2 = \frac{|a|^2}{4} = \frac{1}{4}.$$

Therefore, $r = \frac{1}{2}$. To complete the proof of the theorem, note that by (3.3) we may compute

$$|n-h_a| = \left|\frac{bc}{2a}\right| = \frac{|b||c|}{2|a|} = \frac{1}{2} \quad \text{and}$$
$$\left|n - \frac{a+h}{2}\right| = \left|\frac{a+b+c}{2} - \frac{2a+b+c}{2}\right| = \frac{1}{2}.$$

Similar computations imply that

$$|n-h_b| = |n-h_c| = \left|n - \frac{b+h}{2}\right| = \left|n - \frac{c+h}{2}\right| = \frac{1}{2}.$$

Therefore, all nine points lie on the circle with center $n = \frac{a+b+c}{2}$ and radius $\frac{1}{2}$. □

Chapter 4 Cyclic Quadrilaterals

After triangles, quadrilaterals are the next interesting family of objects to study. Nevertheless, we usually need to make extra assumptions to obtain nice properties. These extra assumptions define specific families of quadrilaterals. In particular, we study the family of cyclic quadrilateral in this section, i.e., quadrilaterals whose vertices all lie on a single circle. If a quadrilateral $\square ABCD$ is cyclic, the cross ratio

$$(a;b;c;d) = \frac{(a-c)(b-d)}{(a-d)(b-c)}$$

is real. We may then assume, after applying translations and homotheties, that the vertices take special forms. For instance, we may assume that the circumcircle of $\square ABCD$ is the unit circle $\mathbb{S}$. If so, the complex numbers a, b, c and d are all on $\mathbb{S}$. This assumption simplifies the arguments significantly. Another useful trick is to assume that one of the vertices, say d, is moved to the origin. Then the cross ratio becomes

$$(a;b;c;0) = \frac{b(a-c)}{a(b-c)}.$$

Ptolemy's Theorem

One of the most important theorems about cyclic quadrilaterals is Ptolemy's theorem. It provides a very useful necessary and sufficient condition for a quadrilateral to be cyclic. This theorem is closely related to the classification of cyclic quadrilateral using the cross ratio.

Theorem 4.1 (Ptolemy's Theorem)

For every convex quadrilateral $\square ABCD$ we have the Ptolemy's inequality:

$$AC \cdot BD \leq AB \cdot CD + AD \cdot BC. \tag{4.1}$$

The convex quadrilateral $\square ABCD$ is cyclic if and only if equality holds in Ptolemy's inequality.

Proof. After applying a translation, we may move D to the origin. The Ptolemy's inequality (4.1) is then satisfied if and only if

$$0 \geq |b-a||c| + |c-b||a| - |c-a||b| = |bc-ac| + |ac-ab| - |bc-ab|,$$

which is precisely the triangle inequality for the triple (bc, ac, ab). Equality occurs precisely when $ac-ab$ is a positive real multiple of $bc-ac$. On the other hand,

$$\frac{bc-ac}{ac-ab} = \frac{(b-a)(c-d)}{(a-d)(c-b)} = -(a;c;b;d).$$

In particular, if $\square ABCD$ is not cyclic, $ac-ab$ cannot be a real multiple of $bc-ac$. If equality is satisfied in Ptolemy's inequality, it follows that $(a;c;b;d)$ is real and $\square ABCD$ is thus cyclic. It is then easy to conclude from the convexity of $\square ABCD$ that $(a;c;b;d)$ is negative. $\square$

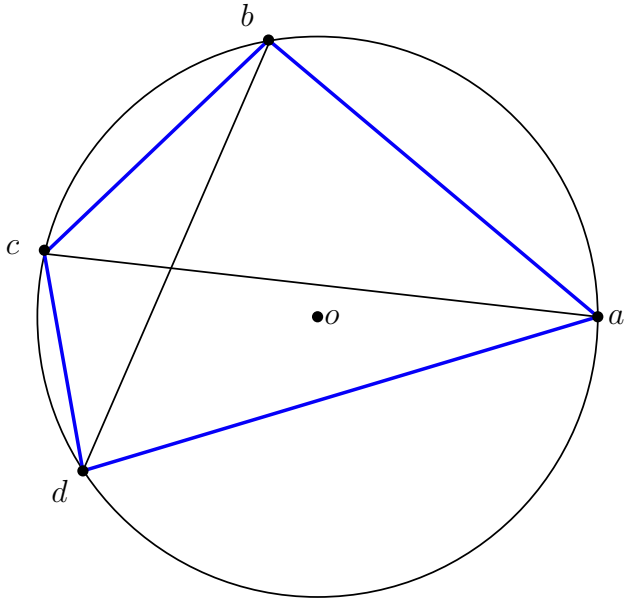


Figure 4.1: The quadrilateral $\square ABCD$ is cyclic if and only if the Ptolemy equality is satisfied: i.e. the sum of the products of opposite side lengths is the product of diagonal lengths.

Japanese Theorem

Let $\square ABCD$ be a cyclic quadrilateral with circumcenter O. Every vertex $V = \{A, B, C, D\}$ determine a triangle. For instance, removing D determines $\triangle ABC$, which has centroid M_d, orthocenter H_d, incenter K_d, and circumcenter O. Similarly, if X is any element in V, by removing X from V we obtain a triangle with centroid M_x, orthocenter H_x, incenter K_x, and circumcenter O. The following theorem gives interesting properties of these centers, including the famous Japanese theorem.

Theorem 4.2

Let $\square ABCD$ be a cyclic quadrilateral. For $x \in \{a, b, c, d\}$ define the centroid M_x, orthocenter H_x, and incenter K_x as discussed above. Then,

- *The quadrilateral $\square H_a H_b H_c H_d$ is congruent to $\square ABCD$;*
- *The quadrilateral $\square M_a M_b M_c M_d$ is cyclic and homothetic to $\square ABCD$;*
- **Japanese theorem:** *The quadrilateral $\square K_a K_b K_c K_d$ is a rectangle.*

Proof. Let us define $\sigma = a + b + c + d$. It then follows that

$$h_x = \sigma - x \quad \text{and} \quad m_x = \frac{\sigma - x}{3} \quad \forall\, x \in \{a, b, c, d\}.$$

Therefore, $\square H_a H_b H_c H_d$ and $\square M_a M_b M_c M_d$ are obtained from $\square ABCD$ by applying the transformations f and g, where

$$f(z) = \sigma - z \quad \text{and} \quad g(z) = \frac{\sigma - z}{3}.$$

Thus, $\square H_a H_b H_c H_d$ is obtained from $\square ABCD$ by a $180°$ rotation followed by a translation, while $\square M_a M_b M_c M_d$ is obtained from $\square ABCD$ by a $180°$ rotation followed by a translation and a homothety

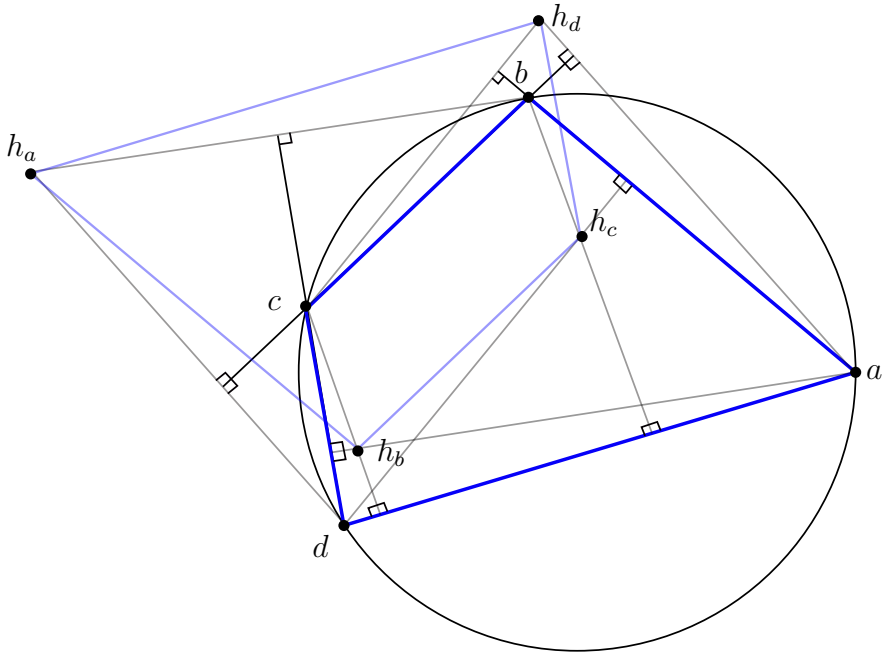


Figure 4.2: H_a, H_b, H_c and H_d are the orthocenters of triangles $\triangle BCD$, $\triangle ACD$, $\triangle ABD$ and $\triangle ABC$, respectively, where $\square ABCD$ is a cyclic quadrilateral. It follows that quadrilaterals $\square H_aH_bH_cH_d$ and $\square ABCD$ are congruent.

with ratio $\frac{1}{3}$. This completes the proof of the first two claims.

For the third part, we must show that

$$R_{k_d,k_a} = -R_{k_a,k_b} = R_{k_b,k_c} = -R_{k_c,k_d}. \tag{4.2}$$

The first equality is equivalent to

$$\lambda := \frac{k_a - k_d}{k_b - k_a} \in i\mathbb{R}.$$

We may further assume that $a = 1$. Let X denote the midpoint of the minor arc $\overset{\frown}{AB}$, Y denote the midpoint of the minor arc $\overset{\frown}{AD}$, U denote the midpoint of the arc $\overset{\frown}{ABC}$, and W denote the midpoint of the arc $\overset{\frown}{ADC}$. Then x is a square root of b, y is a square root of d, while u and $w = -u$ are both square roots of c. Moreover, the midpoints of the minor arcs $\overset{\frown}{BC}$ and $\overset{\frown}{DC}$ correspond to the complex numbers xu and $yw = -yu$, respectively. Further, the midpoint of the arc $\overset{\frown}{BAD}$ corresponds to xy, while the midpoint of the arc $\overset{\frown}{BCD}$ corresponds to $-xy$. With these observations in place, we find

$$k_a = xu - yu + xy, \quad k_d = x + xu - u \quad \text{and} \quad k_b = y - yu + u.$$

Therefore, the ratio λ and its conjugate may be computed:

$$\begin{aligned} \lambda &= \frac{xy - yu - x + u}{y + u - xu - xy} = \frac{(y-1)(x-u)}{(1-x)(y+u)}, \\ \Rightarrow \quad \overline{\lambda} &= \frac{(1-y)(u-x)}{(x-1)(u+y)} = -\lambda. \end{aligned}$$

This implies $\lambda \in i\mathbb{R}$, completing the proof of the first equality in (4.2). The other two equalities are proved similarly. $\square$

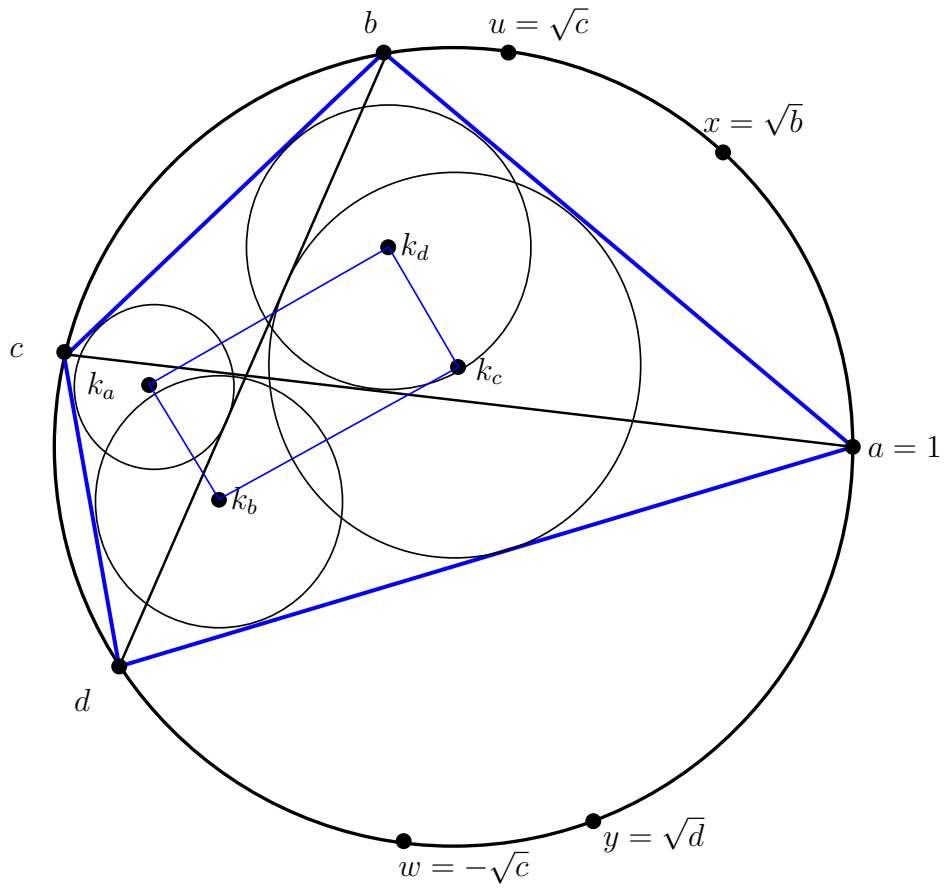


Figure 4.3: K_a, K_b, K_c and K_d are the incenters of triangles $\triangle BCD$, $\triangle ACD$, $\triangle ABD$ and $\triangle ABC$, respectively, where $\square ABCD$ is a cyclic quadrilateral. It follows that quadrilateral $\square K_a K_b K_c K_d$ is a rectangle. This is usually called the *Japanese theorem*.

Chapter 5 IMO Problems and Solutions via Complex Numbers

In this chapter, complex numbers are used to give solutions to IMO geometry problems appearing in contests from 1994-2025.

Problem 2, IMO 1996

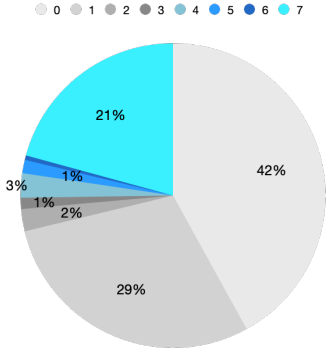
Problem

Let P be a point inside triangle $\triangle ABC$ such that

$$\angle APB - \angle ACB = \angle APC - \angle ABC. \tag{5.1}$$

Let D and E be the incenters of triangles $\triangle APB$ and $\triangle APC$, respectively. Show that AP, BD and CE meet at a point.

Comment 5.1: This problem was proposed by Jeffrey Grossman. A total of 88 complete solutions were received in IMO 1996 for this problem from the 424 participants. The average score was 2.



Solution. Let $\mathbb{S}$ be the circumcircle of $\triangle ABC$. For every $z \in \mathbb{S}$, denote the second intersection of $\mathcal{L}_{p,z}$ with $\mathbb{S}$ by $f(z)$. We may think of f as a function from unit circle to itself:

$$f : \mathbb{S} \to \mathbb{S}, \quad \text{where} \quad f(z) = \frac{z-p}{z\overline{p}-1} \quad \forall\, z \in \mathbb{S}.$$

Assume $f(a) = 1$ and set

$$A' = f(A), \quad B' = f(B) \quad \text{and} \quad C' = f(C).$$

Let M and N be the midpoints of $\overset{\frown}{AB'}$ and $\overset{\frown}{AC'}$, respectively. Then (5.1) implies $A'B' = A'C'$, i.e., $b'c' = 1$. Set $\lambda = \frac{m}{b'}$ and $\mu = \frac{n}{c'}$. Then $a = b'\lambda^2 = c'\mu^2$. Moreover, $b'\frac{\lambda}{\mu} = c'\frac{\mu}{\lambda}$ is the midpoint of $\overset{\frown}{B'AC'}$. Thus

$$\frac{\lambda b'}{\mu} = \frac{\mu c'}{\lambda} = -1 \quad \Rightarrow \quad b' = \frac{1}{c'} = -\frac{\mu}{\lambda} \quad \text{and} \quad a = -\lambda\mu.$$

Moreover, we compute

$$m = -\mu \quad \text{and} \quad n = -\lambda.$$

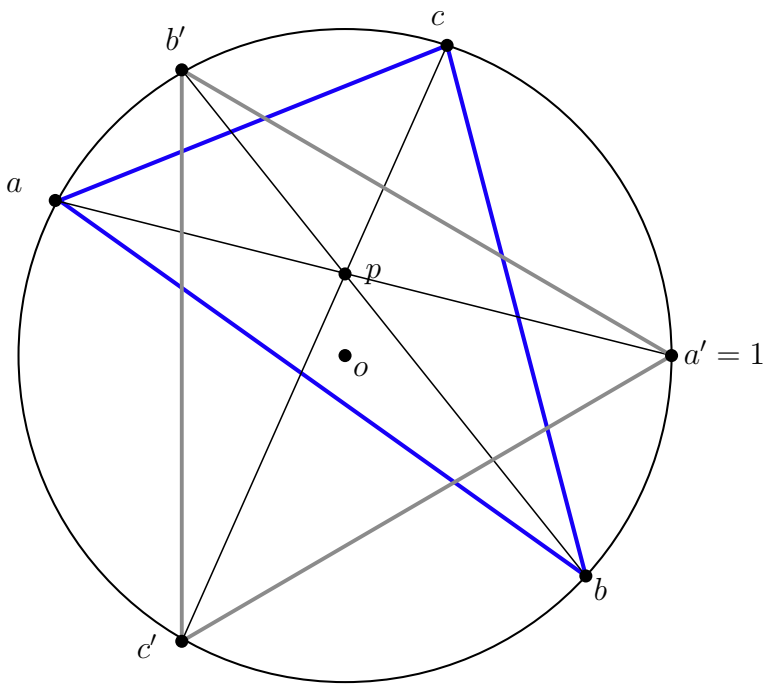


Figure 5.1: Problem 2, IMO 1996.

Since $f(1) = a$, we find $\frac{p-1}{\overline{p}-1} = \lambda\mu$. Setting $q = p - 1$, this reads $q = \lambda\mu\overline{q}$. Moreover, we compute

$$b = f(b') = f\Big(-\frac{\mu}{\lambda}\Big) = \frac{\mu + \lambda p}{\mu\overline{p} + \lambda} \quad \text{and}$$
$$c = f(c') = f\Big(-\frac{\lambda}{\mu}\Big) = \frac{\mu p + \lambda}{\mu + \lambda\overline{p}}.$$

Denote the intersections of $\mathcal{L}_{b,d} = \mathcal{L}_{b,m}$ and $\mathcal{L}_{c,e} = \mathcal{L}_{c,n}$ with line $\mathcal{L}_{a,p} = \mathcal{L}_{a,a'}$ by x and y, respectively. Then x may be computed:

$$x + a\overline{x} = 1 + a \quad \text{and} \quad x - b\mu\overline{x} = b - \mu$$
$$\Rightarrow \quad x = \frac{\mu(b-a) + ab(\mu+1)}{b\mu + a} = \frac{b(1-\lambda) + \lambda\mu(1-b)}{b - \lambda}.$$

From these computations and the equality

$$p = \lambda\mu\overline{p} - \lambda\mu + 1$$

we conclude

$$\begin{aligned} x &= \frac{b(1-\lambda) + \lambda\mu(1-b)}{b-\lambda} \\ &= \frac{(\mu+\lambda p)(1-\lambda) + \lambda\mu((\mu\overline{p}+\lambda) - (\mu+\lambda p))}{(\mu+\lambda p) - \lambda(\mu\overline{p}+\lambda)} \\ &= \frac{(\lambda+\mu) + \lambda\mu(\lambda+\mu+\lambda\mu)(\overline{p}-1)}{(\lambda+\mu) - \lambda\mu(\overline{p}-1)}. \end{aligned}$$

Thus $x = y$, since this expression is symmetric in μ and λ. □

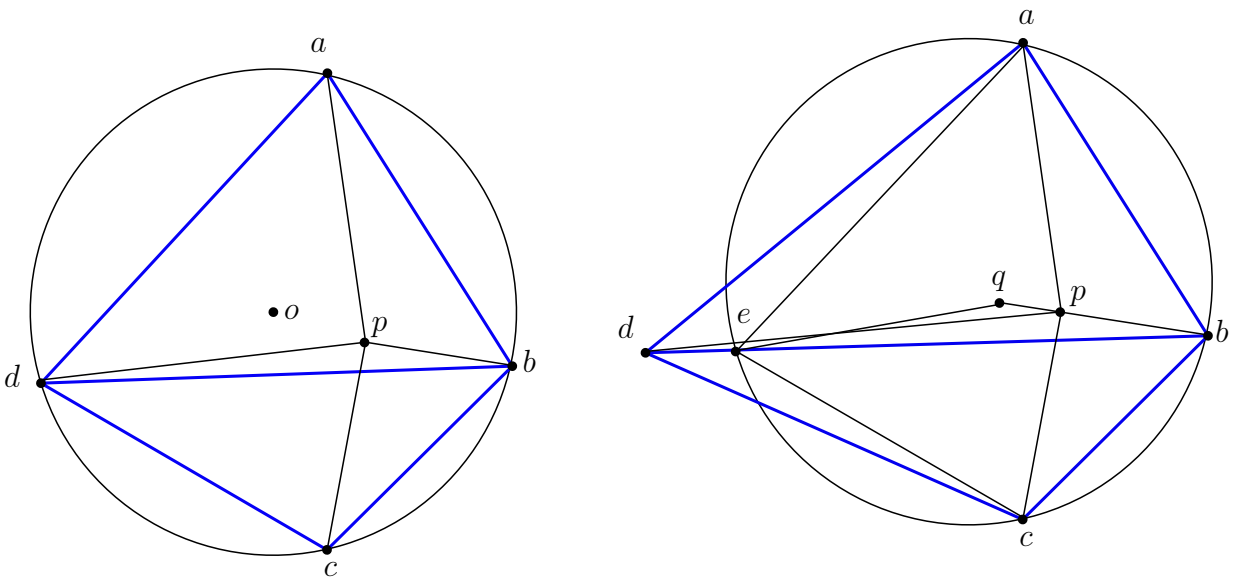


Figure 5.2: Problem 5, IMO 2004.

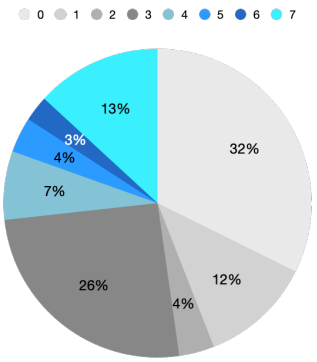
Problem 5, IMO 2004

Problem

In a convex quadrilateral $\square ABCD$, diagonal BD does not bisect angles $\angle ABC$ and $\angle CDA$. Point P lies inside $\square ABCD$ and satisfies

$$\angle PBC = \angle DBA \quad \textit{and} \quad \angle PDC = \angle BDA.$$

Prove that $\square ABCD$ is a cyclic quadrilateral if and only if $AP = CP$.



Comment 5.2: This problem was proposed by Waldemar Pompe. A total of 64 complete solutions were received in IMO 2004 for this problem from the 486 participants. The average score was 2.5.

Solution. First assume $\square ABCD$ is cyclic (see Figure 5.2 left). Therefore $a, b, c, d \in \mathbb{S}$ and we may further assume the perpendicular bisector of AC is $\mathbb{R}$ (i.e., $ac = 1$). Then

$$\begin{aligned}
\angle PBC = \angle DBA \quad &\Rightarrow \quad -\frac{db^2(p-b)}{b\overline{p}-1} = R_{p,b}R_{b,d} = R_{b,c}R_{a,b} = b^2 \\
&\Rightarrow \quad (1) \quad dp + b\overline{p} = 1 + bd \quad \text{and} \\
\angle PDC = \angle BDA \quad &\Rightarrow \quad (2) \quad bp + d\overline{p} = 1 + bd \\
(1), (2) \quad &\Rightarrow \quad p = \overline{p} \quad \text{i.e.,} \quad p \in \mathbb{R}.
\end{aligned}$$

Thus P is on the perpendicular bisector of AC. Now assume D is not on the circumcircle $\Omega = \mathbb{S}$ of

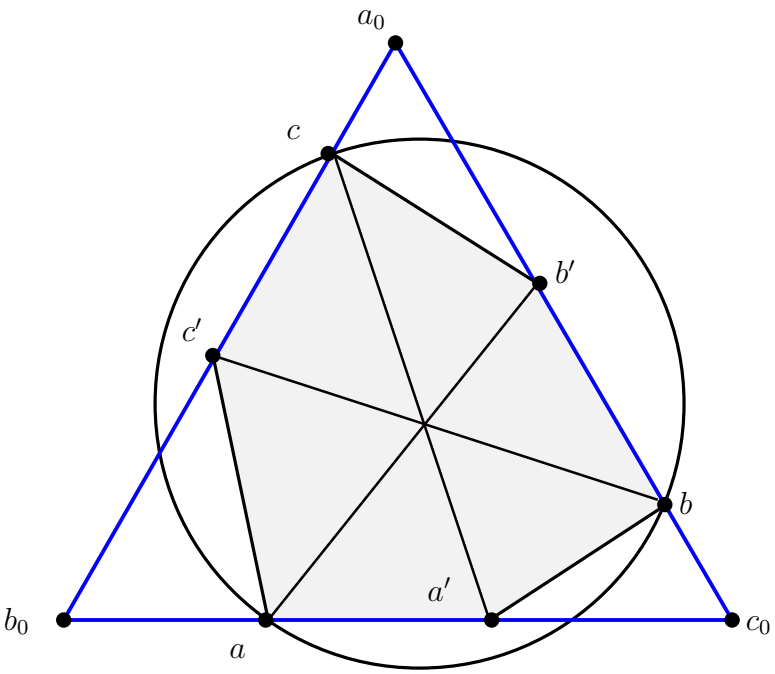


Figure 5.3: Problem 1, IMO 2005.

$\triangle ABC$ (see Figure 5.2 right). Again, assume the perpendicular bisector of AC is the real line. Let BD intersect Ω at E, and let Q be a point on BP such that

$$\angle BEA = \angle QEC.$$

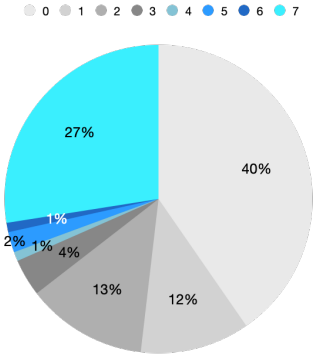
It is not hard to see $P \neq Q$. However, the first part implies Q is on the perpendicular bisector of AC. Since P is also on this bisector, BP is the perpendicular bisector of AC. But then BP is a segment on line BD, implying BD bisects $\angle ABC$, contradicting our initial assumption. □

Problem 1, IMO 2005

Problem

Given equilateral triangle $\triangle A_0B_0C_0$, points A, A' on B_0C_0, B, B' on C_0A_0 and C, C' on A_0B_0 are chosen such that they are vertices of a convex hexagon $AA'BB'CC'$ with equal side lengths. Prove that lines AB', BC' and CA' are concurrent.

Comment 5.3: This problem was proposed by Bogdan Enescu. A total of 141 complete solutions were received in IMO 2005 for this problem from the 513 participants. The average score was 2.6.



Solution. Assume the circumcircle of $\triangle ABC$ is $\mathbb{S}$, while

$$a' = a + d, \quad b' = b + \lambda d \quad \text{and} \quad c' = c + \lambda^2 d,$$

where $d \in \mathbb{R}$ and $\lambda = e^{2\pi i/3}$. Since

$$AA' = A'B, \quad BB' = B'C \quad \text{and} \quad CC' = C'A,$$

A' is on the perpendicular bisector $\mathcal{L}_c$ of AB, B' is on $\mathcal{L}_a$ of BC, and C' is on $\mathcal{L}_b$ of CA. Therefore,

$$\frac{1}{a} + d = \overline{a}' = \frac{a'}{ab} = \frac{a+d}{ab} \quad \Rightarrow \quad d = \frac{b-a}{1-ab}.$$

Applying the same argument for b' and c' we find that in fact

$$d = \frac{b-a}{1-ab} = \frac{c-b}{\lambda - \lambda^2 bc} = \frac{a-c}{\lambda^2 - \lambda ca}.$$

Equating the first two expressions for d gives:

$$\frac{b-a}{1-ab} = \frac{c-b}{\lambda - \lambda^2 bc} \quad \Rightarrow \quad c = -\frac{\lambda a + \lambda^2 b + ab^2}{1 + \lambda ab + \lambda^2 b^2}.$$

Replacing in the equality between the first and the third expression we find

$$\begin{aligned}
&\Rightarrow \quad \frac{b-a}{1-ab} = \frac{a-c}{\lambda^2 - \lambda ca} = \frac{(b-a)(\lambda^2 - \lambda ab)}{\lambda^2 + 2ab + \lambda b^2 + \lambda^2 a^2 + \lambda a^2 b^2} \\
&\Rightarrow \quad (1-ab)(\lambda^2 - \lambda ab) = \lambda^2 + 2ab + \lambda b^2 + \lambda^2 a^2 + \lambda a^2 b^2 \\
&\Rightarrow \quad 0 = ab + \lambda b^2 + \lambda^2 a^2 = \lambda (b-a)(b - \lambda a) \\
&\Rightarrow \quad b = \lambda a.
\end{aligned}$$

Similarly, or from the computation of c in terms of a and b, we have $c = \lambda b = \lambda^2 a$, i.e., $\triangle ABC$ is equilateral. Therefore, A and B' are both on the perpendicular bisector $\mathcal{L}_a$, which means $\mathcal{L}_a = \mathcal{L}_{a,b'}$. Similarly, $\mathcal{L}_b = \mathcal{L}_{b,c'}$ and $\mathcal{L}_c = \mathcal{L}_{c,a'}$. In particular, $\mathcal{L}_{a,b'}$, $\mathcal{L}_{b,c'}$ and $\mathcal{L}_{c,a'}$ all pass through the origin. □

Problem 2, IMO 2010

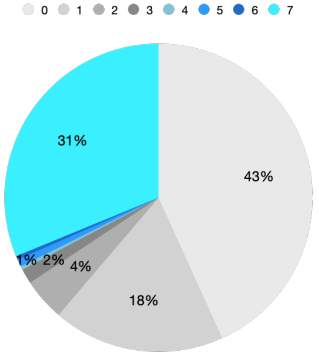
Problem

Let K be the incentre of triangle $\triangle ABC$ and let Γ be its circumcircle. Let line AK intersect Γ again at D. Let E be a point on arc $\overset{\frown}{BDC}$ and F a point on side BC such that

$$\angle BAF = \angle CAE < \frac{1}{2}\angle BAC.$$

Finally, let G be the midpoint of segment KF. Prove that lines DG and EK intersect on Γ.

Comment 5.4: This problem was proposed by Tai Wai Ming and Wang Chongli. A total of 161 complete solutions were received in IMO 2010 for this problem from the 516 participants. The average score was 2.6.



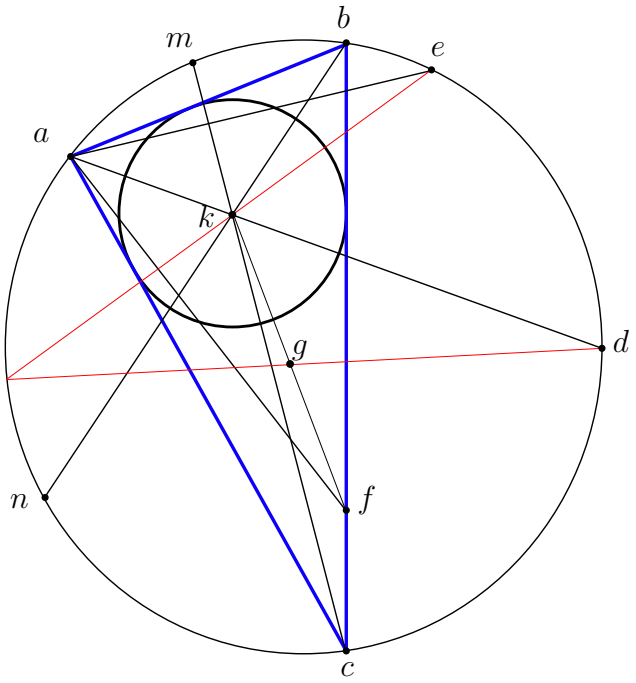


Figure 5.4: Problem 2, IMO 2010.

Solution. Assume Γ is the unit circle and $d = 1$. Denote midpoints of minor arcs $\overset{\frown}{AB}$ and $\overset{\frown}{AC}$ by M and N, respectively. Then

$$b = -\frac{m}{n}, \quad c = -\frac{n}{m}, \quad a = -mn \quad \text{and} \quad k = m + n + 1.$$

Since $f \in \mathcal{L}_{b,c}$, we have

$$f + \overline{f} = -\frac{m^2 + n^2}{mn}.$$

The assumption implies

$$\frac{f + mn}{1 - mnf - m^2 - n^2} = \frac{f - a}{mn(\overline{f} - \overline{a})} = \frac{R_{a,f}}{mn} = \frac{R_{a,c}R_{a,b}}{mnR_{a,e}} = -\frac{a^2bc}{a^2e} = \frac{1}{e}.$$

Therefore, we may compute e and g:

$$e = \frac{1 - m^2 - n^2 - mnf}{f + mn} \quad \text{and} \quad g = \frac{f + m + n + 1}{2}.$$

The intersection points $z \in \mathcal{L}_{d,g}$ and $w \in \mathcal{L}_{e,k} \cap \Gamma$ are:

$$\begin{aligned} w &= \frac{g - 1}{1 - \overline{g}} = \frac{mn(m + n + f - 1)}{mn(1 + f) + m(m - 1) + n(n - 1)} \quad \text{and} \\ z &= \frac{k - e}{1 - e\overline{k}} \\ &= \frac{mn\Big((m + n + 1)(f + mn) + m^2 + n^2 + mnf - 1\Big)}{mn(f + mn) + (mn + m + n)(mnf + m^2 + n^2 - 1)} \\ &= \frac{mn(m + 1)(n + 1)(f + m + n - 1)}{(m + 1)(m^2 - 1)(n^2 - 1) - (n + 1)(mnf + m^2 + n^2 - 1)} = w. \end{aligned}$$

Therefore, lines DG and EK intersect Γ at the same point. □